1.4.8 Subset

Let A and B be two sets. If every element of A is also an element of B, then A is called a subset of B. We write $A \subseteq B$.

$A \subseteq B$ is read as "A is a subset of B"

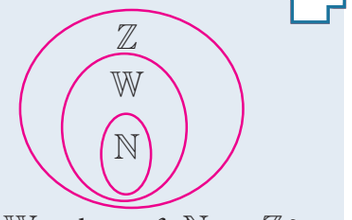
Thus $A \subseteq B$, if $a \in A$ implies $a \in B$.

If A is not a subset of B, we write $A \not\subseteq B$

Clearly, if A is a subset of B, then $n(A) \leq n(B)$.

Since every element of A is also an element of B, the set B must have at least as many elements as A, thus $n(A) \leq n(B)$.

Thinking Corner



Is $\mathbb{W}$ subset of $\mathbb{N}$ or $\mathbb{Z}$?

The other way is also true. Suppose that $n(A) > n(B)$, then A has more elements than B, and hence there is at least one element in A that cannot be in B, so A is not a subset of B.

For example,

(i) $\{1\} \subseteq \{1,2,3\}$ (ii) $\{2,4\} \not\subseteq \{1,2,3\}$

Example 1.6

Insert the appropriate symbol $\subseteq$ or $\not\subseteq$ in each blank to make a true statement. (i) $\{10, 20, 30\}$ _____ $\{10, 20, 30, 40\}$ (ii) $\{p, q, r\}$ _____ $\{w, x, y, z\}$

Solution

(i) $\{10, 20, 30\}$ _____ $\{10, 20, 30, 40\}$

Since every element of $\{10, 20, 30\}$ is also an element of $\{10, 20, 30, 40\}$, we get $\{10, 20, 30\} \subseteq \{10, 20, 30, 40\}$.

(ii) $\{p, q, r\}$ _____ $\{w, x, y, z\}$

Since the element p belongs to $\{p, q, r\}$ but does not belong to $\{w, x, y, z\}$, shows that $\{p, q, r\} \not\subseteq \{w, x, y, z\}$.

Activity-3

Discuss with your friends and give examples of subsets of sets from your daily life situation.

Example 1.7

Write all the subsets of $A = \{a, b\}$.

Solution

$A = \{a,b\}$

Subsets of A are $\varnothing, \{a\}, \{b\}$ and $\{a, b\}$.

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